

Exports, parser capabilities, and progressive IR
stages

Export targets

The primary export targets are the **Generalized Notation Notation (GNN)** canonical Markdown (`model.gnn.md`) and the equivalent companion JSON files described in `../cogant/docs/export/README.md`. Optional interop targets (GraphML, Parquet) support analysis in Gephi/yEd and DuckDB, and optional tensor views for PyTorch Geometric, DGL, or HDF5 can be selected when downstream graph neural network training pipelines need to consume the program graph as a relational tensor. Ensure the Python environment includes optional dependencies for these tensor exports when those code paths are used.

Python AST parser capabilities

The v0.5.x front end relies on `cogant.static.parser.PythonASTParser`, which processes Python source through the standard-library `ast` module at the CPython version available in the runtime (3.11+ required, consistent with the `requires-python = ">=3.11"` declared in `../cogant/pyproject.toml`). The parser extracts the following construct categories:

- ▶ **Module-level entities:** module docstrings, `__all__` exports, top-level assignments.
- ▶ **Functions and methods:** `def` and `async def`, including signatures with positional, keyword, variadic (`*args`, `**kwargs`), and positional-only parameters. Default values are recorded as constant expressions where statically evaluable.
- ▶ **Classes:** class definitions, base classes, metaclasses, and the `__init__` / `__new__` boundary.
- ▶ **Decorators:** `@staticmethod`, `@classmethod`, `@property`, `@dataclass`, and arbitrary user-defined decorators. Decorator arguments are captured as attribute metadata

Progressive IR stages

Processing advances through six intermediate representations, each adding semantic detail atop its predecessor. Table 3 summarizes what each stage contributes.

Table 3. Progressive IR stages and their contributions.

Stage	IR name	Key additions	Typical output size (10K-function repo)
1	Repo IR	Raw entities and relationships per file; deduplication; merged type info	~15 MB JSON
2	Program Graph IR	Consolidated directed graph $G =$	~20 MB JSON